

OGTech Internship Program 2026

Hello Team,

Welcome to OGTech! We are thrilled to have you join us for our Internship Program.

Over the next month, you will work together as one team to build a real-world software project from scratch. The goal is to gain practical, hands-on experience across all areas of software development — from understanding a business problem, to designing a system, to delivering a working product.

Our project is called "**NurseryLink**".

Project Overview: NurseryLink

NurseryLink is a web application designed to manage nursery operations and improve communication between the nursery and parents.

The application has three types of users:

- **Admin** — manages teachers, students, parents, and class assignments. Has a high-level view of the nursery and can create additional admin accounts with specific privileges.
- **Teachers** — record daily activities for the children in their assigned class.
- **Parents** — view information related to their own children only.

Platform: The primary deliverable is a **web application**. A mobile application is **optional** and can be built as a bonus if the team finishes the core requirements ahead of schedule.

What the System Must Do

The following sections describe the business requirements of the system. It is your responsibility as a team to analyze these requirements, ask the right questions, and design the system accordingly.

1. Accounts & Access Control

The system does not allow self-registration. All accounts are created by an Admin. Every user in the system — whether Admin, Teacher, or Parent — has a **full name**, a **unique username**, an **email address** used for login, and a **password** that must never be stored in plain text. Each account also carries its **role** (Admin, Teacher, or Parent) and an **active/inactive status**, since accounts can be deactivated without being deleted.

A Teacher can only interact with students in their own assigned class. A Parent can only see information about their own children. The system must strictly enforce these boundaries — no user should ever be able to access or modify data outside their permitted scope.

There is one default Admin account that exists when the system is first set up. This account cannot be deactivated.

2. Admin Privileges

An Admin can create additional Admin accounts. However, not all admins have the same level of access. When creating a new Admin account, the creating Admin selects a specific set of privileges to assign to it. Each privilege corresponds to one of the available admin capabilities:

- Managing teacher accounts.
- Managing student records and class assignments.
- Managing parent accounts and their links to students.
- Viewing reports and the high-level dashboard.
- Managing other admin accounts.

The system must record which privileges each Admin account holds, and must know which Admin created each Admin account. An Admin cannot grant a privilege they do not themselves hold. The default Admin account holds all privileges and is the only one guaranteed to always have full access.

3. Teacher Management

An Admin with the appropriate privilege can create teacher accounts and deactivate them when a teacher leaves the nursery. Beyond the common account fields (name, username, email, password), no additional information is required for a teacher beyond their class assignment, which is managed separately.

A deactivated teacher cannot log in, but all records they created must remain intact and unchanged in the system.

4. Classes & Student Management

The nursery is organized into named classes. Each class has a **class name** and is assigned to exactly one teacher at a time. An Admin with the appropriate privilege manages all class and student records.

Each student has a **full name**, a **date of birth**, and a **unique student code** that identifies them across the system. A student belongs to one class at a time. The system must always know which class a student is currently assigned to.

When a student needs to be transferred to a different class, the Admin performs this change. The transfer only affects future activity — all historical records must remain associated with the original class context in which they were created.

If a class needs to be reassigned to a different teacher, the Admin handles this. The system must support this transition without any loss of data.

5. Parent Management

An Admin with the appropriate privilege manages all parent accounts. Beyond the common account fields, no additional information is required for a parent.

Each parent must be linked to one or more students. The system must know which parents are connected to which students, as this determines what each parent can see and which parents receive notifications. A student can have more than one parent linked to their profile.

The Admin can update these links at any time — adding a new parent to a student or removing an existing link when no longer applicable.

6. High-Level Reports & Dashboard

An Admin with the appropriate privilege has access to a high-level view of the nursery. This is not a detailed activity log — it is a summary view intended to surface things that require attention across all classes.

What counts as "requiring attention" is something your team should think through based on the features in this document. The goal is to give the Admin a meaningful, at-a-glance picture of what is happening across the nursery without having to dig into individual records.

7. Daily Meals Log

Throughout the day, children have set meal times: **Breakfast**, **Lunch**, and **Snack**. After each meal, the teacher records the **meal type** and the **eating status** — whether the child ate everything, ate only some of it, or refused to eat. The system must also record **who logged the entry** and the **exact date and time** it was recorded.

The system should prevent duplicate or illogical entries — it must not be possible to log the same meal type twice for the same child on the same day.

Parents can view this information.

8. Toilet Visits Log

Teachers log each time a child uses the toilet. Each entry records the **type of visit** (whether it was a pee or a poo), an **optional free-text note** from the teacher, **who logged it**, and the **date and time** of the visit. There is no limit on how many visits can be logged per child per day.

Parents can view this information.

9. Temperature Checks

Teachers periodically check children's temperatures and record each reading. Each entry captures the **temperature value in Celsius**, an **optional free-text note**, **who recorded it**, and the **date and time** of the reading. A valid reading must fall within a realistic human body temperature range — values outside this range should not be accepted.

If a reading indicates a fever, the system must automatically notify all parents linked to that child without any manual action from the teacher.

Parents can view their child's full temperature history.

10. Notifications

Parents receive automatic notifications from the system when important events occur. Each notification is associated with a **specific parent**, a **specific student**, and carries a **type** that identifies what triggered it, a **message** describing what happened, a **read/unread status**, and the **date and time** it was created.

A parent can view their notifications and mark them as read. Notifications are never created manually — they are always triggered by something that happens in the system.

Extra Features

The following features are more complex. They are not required for the core delivery but are expected from teams that want to challenge themselves. Some of them intentionally introduce complications to the data model, business logic, and access control that you will need to think through carefully.

Extra 1: Attendance Tracking

At the start of each day, the teacher marks each student in their class as either present or absent. Each attendance entry records the **student**, the **date**, the **status** (present or absent), an **optional reason** in case the absence was communicated in advance, and **who recorded it**.

If a student is marked absent, the system should prevent any activity logs (meals, toilet visits, temperature checks) from being recorded for that student on that day.

If a student accumulates 3 or more consecutive absent days, the system should automatically alert all parents linked to that student. Admins with dashboard access can see an attendance summary across all classes.

Extra 2: Incident Reports

Teachers can file an incident report when something notable happens to a student — a fall, a conflict, an allergic reaction, or similar. Each report captures the **student involved**, a **description** of what happened, a **severity level**, the **date and time** of the incident, and **who filed the report**.

When a report is filed, all parents linked to that student must be notified and must explicitly acknowledge the report. The system must track the **acknowledgment status** per parent — recording whether each parent has acknowledged and **when** they did so.

The teacher can see which parents have and have not acknowledged. Admins with dashboard access can view all incident reports across the nursery and their full acknowledgment status.

Extra 3: Meal Pattern Alert

If a student has refused to eat at all across an entire day for 3 or more consecutive days, the system must automatically notify the parents linked to that student and flag the student on the teacher's view.

This requires the system to look back at historical meal data, not just today's records. Your team must decide how to handle days where some meals were never logged at all.

Extra 4: End-of-Day Summary

At the end of each school day, the system makes a daily summary available to parents for each of their children. The summary consolidates all activity logs recorded for that student on that day into a single readable view — meals, toilet visits, temperature readings, and any incidents.

The system should determine on its own when the day is considered complete for a student. Think carefully about what conditions would make a day complete, and what should happen if a teacher never finishes logging.

Extra 5: Class Announcements

A teacher can post an announcement visible to all parents of students in their class (e.g., "Bring a costume on Thursday", "Field trip permission slip due Friday"). Each announcement has a **title**, a **body text**, the **class it belongs to**, **who posted it**, the **date it was posted**, and an **expiry date**.

After the expiry date passes, the announcement must no longer appear in the active feed but must remain in the system for record-keeping purposes. Parents see only announcements relevant to their children's class.

Extra 6: Supply Requests

A teacher can send a supply request to the parents of a specific student. Each request records the **student it concerns**, the **item being requested**, the **quantity needed**, an **optional note**, **who submitted the request**, and the **date and time** it was submitted.

Parents linked to that student receive a notification and must acknowledge the request. The system must track **whether each parent has acknowledged** the request and **when**. The teacher can see which requests are still pending acknowledgment.

Extra 7: Substitute Teacher Coverage

In some cases, a different teacher needs to temporarily cover a class for one or more days. The system must support recording a **substitute assignment** that captures the **class being covered**, the **teacher assigned as substitute**, the **date or date range** of the coverage, and **which Admin arranged it**.

During the coverage period, the substitute teacher gains access to that class exactly as the assigned teacher would. All activity logs recorded during this period must capture **who actually recorded them**, not just the class's permanent teacher.

Think carefully about how this interacts with access control, and what a substitute should and should not be permitted to do compared to the permanently assigned teacher. This is arranged by an Admin with teacher management privileges.

Required Knowledge Before Starting

To successfully contribute to this project, each participant should be comfortable with the following:

- **Git:** Branching, committing, pushing, and working collaboratively on a shared repository.

- **Object-Oriented Programming:** Classes, inheritance, encapsulation, and how to model real-world concepts in code.
 - **C# Fundamentals:** Classes, methods, data types, and collections.
 - **ASP.NET Core:** How a web API is structured and how HTTP requests are handled.
 - **Entity Framework Core:** Defining models, configuring relationships, and querying data.
 - **Dependency Injection:** How services are registered and used in ASP.NET Core.
 - **SQL Basics:** Tables, relationships, and basic queries.
 - **HTML/CSS & JavaScript basics** (for the web frontend).
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Required Software Installation

Please have the following tools installed before the program begins on **April 6, 2026**:

- **Visual Studio 2022** with ".NET and Web Development" workloads.
- **.NET 8 SDK**.
- **Git** for version control.
- **Postman** for API testing.
- **SQL Server Management Studio (SSMS)** for database management.
- **Flutter SDK** (only if the team chooses to build the optional mobile app).

If you face any issues during installation, please contact the team for assistance before the start date.

Project Benefits

This project will help you:

- Practice translating business requirements into technical decisions.
 - Experience working as part of a real development team.
 - Build and integrate a full-stack web application.
 - Develop judgment around system design, data modeling, and edge cases.
 - Learn to work with and critically evaluate AI-assisted development tools.
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We look forward to seeing how you approach the problem, the decisions you make, and what you build.

If you have any questions, please don't hesitate to ask.

Best regards,
OGTech Team